

1. Python Code to create a csv file, with 5000 rows; subscriptions.csv

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UN PICO 5.09 File: generate_data.py Modified

import random
from datetime import date, timedelta

random.seed(42)

plan_tiers = ["Basic", "Standard", "Premium"]
regions = ["North America", "Europe", "APAC", "LATAM"]
channels = ["Organic", "Paid Ads", "Referral", "Partnership"]
plan_price = {"Basic": 20, "Standard": 50, "Premium": 100}

start_range = date(2024, 1, 1)

file = open("subscriptions.csv", "w")
file.write("customer_id,plan_tier,start_date,end_date,mrr,region,acquisition_channel,downgrade_flag\n")

for i in range(5000):
    customer_id = "CUST" + str(i + 1)

    days_after_start = random.randint(0, 500)
    start_date = start_range + timedelta(days=days_after_start)

    tenure_months = random.randint(1, 24)
    end_date = start_date + timedelta(days=tenure_months * 30)

    plan_tier = random.choice(plan_tiers)
    mrr = plan_price[plan_tier]
    region = random.choice(regions)
    channel = random.choice(channels)
    downgrade_flag = random.choice([0, 0, 0, 1])

    line = str(customer_id) + "," + str(plan_tier) + "," + str(start_date) + "," + str(end_date) + "," + str(mrr) + "," + str(region) + "," + str(channel) + "," + str(downgrade_flag) + "\n"
    file.write(line)

file.close()

print("Done! subscriptions.csv created with 5000 rows")
```

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Cut Text UnCut Text Cur Pos To Spell

2. Uploaded the csv into Tableau

The screenshot shows the Tableau interface with the 'subscriptions.csv' data source loaded. The sidebar on the left contains the 'Connections' section with 'subscriptions' (Text file) and the 'Files' section with 'subscriptions.csv'. The central workspace shows 'subscriptions.csv' with two interactive buttons: 'Relate tables' (Drag tables near subscriptions.csv to relate them) and 'Add a base table' (Drag out additional base tables for multi-fact analysis). Below these buttons is a link 'Need help adding data? Learn more'. The bottom pane displays a data table with 8 fields and 5000 rows. The table has columns: Customer Id, Plan Tier, Start Date, End Date, Mrr, Region, and Acquisition. The data is organized into 8 rows (CUST1 to CUST8) with various values for each field.

subscriptions.csv

8 fields 5000 rows

100 rows

Name subscriptions.csv

Description No description available.

Fields

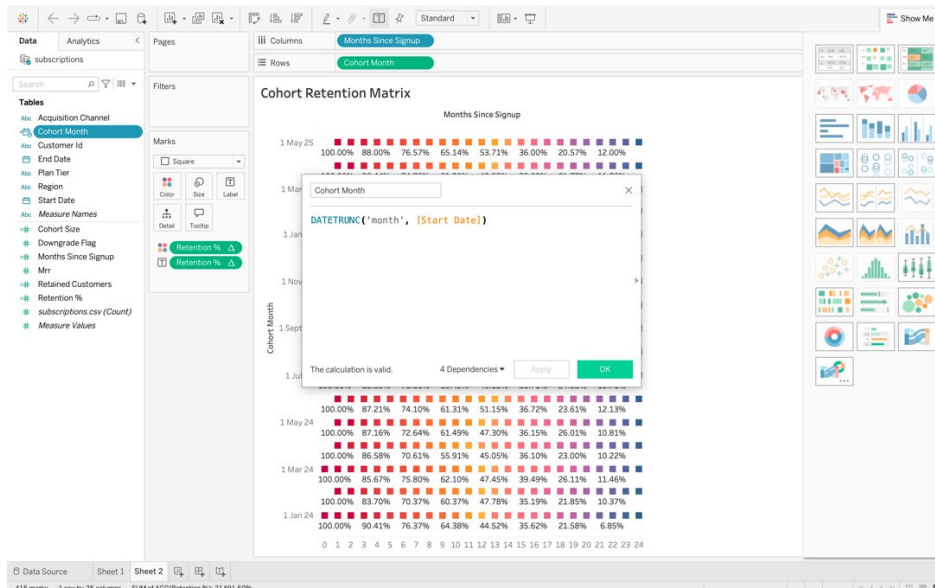
Type	Field Name	Physical Table	Remot...
Abc	Customer Id	subscriptions.csv	custome...
Abc	Plan Tier	subscriptions.csv	plan_tier
Abc	Start Date	subscriptions.csv	start_date
Abc	End Date	subscriptions.csv	end_date
#	Mrr	subscriptions.csv	mrr

Go to Worksheet

Data Source Sheet 1

Customer Id	Plan Tier	Start Date	End Date	Mrr	Region	Acquisition
CUST1	Basic	23/11/2024	23/03/2025	20	APAC	Paid Ads
CUST2	Basic	12/03/2024	02/03/2026	20	North America	Partnershi
CUST3	Basic	16/01/2024	15/04/2024	20	Europe	Organic
CUST4	Premium	01/01/2025	23/09/2026	100	LATAM	Paid Ads
CUST5	Basic	28/10/2024	25/07/2025	20	Europe	Partnershi
CUST6	Basic	22/05/2024	19/10/2024	20	APAC	Organic
CUST7	Standard	13/07/2024	10/11/2024	50	APAC	Referral
CUST8	Premium	08/01/2025	03/04/2026	100	North America	Partnershi

3. Building the Cohort Retention Matrix in Tableau
4. I connected Tableau to my subscriptions.csv file using the Text file connector, then confirmed all 8 columns loaded correctly and that start_date and end_date were recognized as proper date fields.
5. I created a calculated field called Cohort Month using DATETRUNC('month', [Start Date]) to round each customer's signup date down to their signup month, which defines their cohort.
6. Next, I created a field called Months Since Signup using DATEDIFF('month', [Cohort Month], DATETRUNC('month', [End Date])), which calculates how many months each customer was active for, based on their start and end dates.
7. I then created Cohort Size using an LOD expression, { FIXED [Cohort Month] : COUNTD([Customer Id]) }, to count how many customers belonged to each signup cohort.
8. For the retention logic itself, I initially wrote Retained Customers as an LOD comparison, but found it returned 100% for every cell since it was comparing each customer's tenure against itself. I corrected this using a table calculation instead, WINDOW_SUM(COUNTD([Customer Id]), 0, LAST()), which counts customers cumulatively from a given month-since-signup value onward.
9. I updated Retention % to divide that running count by the cohort size: WINDOW_SUM(COUNTD([Customer Id]), 0, LAST()) / SUM([Cohort Size]).
10. To build the matrix, I placed Cohort Month on Rows (set to Exact Date) and Months Since Signup on Columns (set to Discrete and Dimension). I set the mark type to Square, then placed Retention % on both Color and Label.
11. Since Retention % is a table calculation, I edited its "Compute Using" setting to Specific Dimensions, checking only Months Since Signup, so it calculated correctly across each row.
12. Finally, I formatted the Retention % labels as percentages with two decimal places, titled the sheet "Cohort Retention Matrix," and exported a screenshot along with saving the workbook as a .twbx file.



13.

14. RESULTS:

Cohort Retention Matrix

